

ZeroMQ using Java with Maven in NetBeans.

ZeroMQ (ZMQ) is a high-performance messaging library that simplifies building distributed or concurrent applications. We'll set up a basic **Request/Reply** model where one component acts as the client (requester) and another as the server (responder).

Prerequisites:

1. **NetBeans IDE** (ensure **Java 8+** and **Maven** are configured).
2. **ZeroMQ** and its **Java binding (JeroMQ)**.

Step 1: Create a Maven Project in NetBeans

1. Open **NetBeans IDE**.
2. Go to **File** → **New Project**.
3. Choose **Maven** → **Java Application** and click **Next**.
4. Fill in the project details:
 - **Group ID:** `com.example`
 - **Artifact ID:** `zeromq-helloworld`
 - **Version:** `1.0-SNAPSHOT`
5. Click **Finish**.

Step 2: Add JeroMQ Dependency to `pom.xml`

ZeroMQ has a Java binding called **JeroMQ** that provides ZeroMQ capabilities for Java.

Open the `pom.xml` file and add the following dependency to use **JeroMQ**:

```
<dependencies>
  <!-- JeroMQ (ZeroMQ Java binding) -->
  <dependency>
    <groupId>org.zeromq</groupId>
    <artifactId>jeromq</artifactId>
    <version>0.5.2</version> <!-- Ensure to check for the latest version -->
  </dependency>
</dependencies>
```

Step 3: Implement the ZeroMQ Server (Responder)

1. Right-click on the `src/main/java` directory, then create a new Java class called `ZmqServer` inside the `com.example` package.
2. Add the following code to implement the server, which will receive and reply to messages from the client:

```
package com.example;
```

```

import org.zeromq.ZMQ;

public class ZmqServer {

    public static void main(String[] args) {
        // Create a ZeroMQ context
        ZMQ.Context context = ZMQ.context(1);

        // Create a socket to respond to requests
        ZMQ.Socket responder = context.socket(ZMQ.REP);
        responder.bind("tcp://*:5555");

        System.out.println("ZeroMQ Server started, waiting for requests...");

        while (true) {
            // Wait for the next request from the client
            String request = responder.recvStr(0);
            System.out.println("Received request: " + request);

            // Do some 'work' and send a reply
            String reply = "Hello, Client!";
            responder.send(reply.getBytes(ZMQ.CHARSET), 0);
            System.out.println("Sent reply: " + reply);
        }
    }
}

```

Step 4: Implement the ZeroMQ Client (Requester)

1. Right-click on `src/main/java` again and create a class called `ZmqClient` in the `com.example` package.
2. Add the following code to implement the client, which will send a message to the server and wait for a reply:

```

package com.example;

import org.zeromq.ZMQ;

public class ZmqClient {

    public static void main(String[] args) {
        // Create a ZeroMQ context
        ZMQ.Context context = ZMQ.context(1);

        // Create a socket to send requests
        ZMQ.Socket requester = context.socket(ZMQ.REQ);
        requester.connect("tcp://localhost:5555");

        System.out.println("ZeroMQ Client started, sending request...");

        // Send a request to the server
        String request = "Hello, Server!";
        requester.send(request.getBytes(ZMQ.CHARSET), 0);
    }
}

```

```

        System.out.println("Sent request: " + request);

        // Receive the reply from the server
        String reply = requester.recvStr(0);
        System.out.println("Received reply: " + reply);

        // Close the requester socket and context
        requester.close();
        context.close();
    }
}

```

Step 5: Running the Example

Now that the **ZeroMQ Server** and **Client** are implemented, you can run them.

1. Start the ZeroMQ Server:

1. In **NetBeans**, right-click on the `ZmqServer` class and choose **Run**.
2. The server will start and wait for incoming requests. 2. Run the ZeroMQ Client:
1. In a new terminal or a different instance of **NetBeans**, right-click on the `ZmqClient` class and choose **Run**.
2. You should see the client sending a request to the server and receiving a reply.

Explanation of Code:

- **ZmqServer:**
 - It binds to port 5555 and waits for incoming client requests.
 - Once a request is received, it processes it and replies with a message ("Hello, Client!").
- **ZmqClient:**
 - It connects to the server on port 5555 and sends a request ("Hello, Server!").
 - After sending the request, it waits for the server's reply.